To bypass the thermalization-time trade-off, the field has pivoted away from the "wait-and-read" equilibrium paradigm. Here are the leading strategies and the specific metrological resources they leverage to achieve fast and highly sensitive temperature estimations.

1. Non-Equilibrium and Transient Thermometry

Instead of waiting for the asymptotic steady state ρ_{eq}, these strategies extract thermal information during the transient dynamical evolution of the probe $\rho(t)$ under the Liouvillian.

- **Quantum Coherence as a Resource:** If a probe is initialized in a coherent superposition rather than a classical statistical mixture, the transient QFI can temporarily overshoot the asymptotic equilibrium QFI. The coherence generates oscillations that are highly sensitive to the dissipative parameters of the bath, allowing for a fast, early-time measurement before coherence is entirely lost to decoherence.

- **The Metrological Mpemba Effect:** A very recent development leverages anomalous relaxation (the Mpemba effect, where a hotter system cools faster than a colder one). By preparing the probe in specific non-equilibrium "hot" initial states that project strongly onto the fast-decaying eigenmodes of the Liouvillian, researchers have shown that the transient QFI can be significantly enhanced. Here, the metrological resource is the *anomalous relaxation spectrum* itself, utilizing the fast eigenvalues to circumvent the slow thermalization bottleneck.

2. Many-Body Systems and Criticality

Rather than using single, highly degenerate isolated probes, researchers are turning to interacting many-body quantum systems.

- **Quantum Phase Transitions (QPTs):** At a QPT (e.g., in the Lipkin-Meshkov-Glick, Ising, or XXZ models), the energy gap closes and the system's susceptibility diverges. By operating exactly at or very close to criticality, the macroscopic ground-state properties become hypersensitive to minuscule thermal fluctuations. The metrological resource here is the **divergent correlation length / criticality**. Because the entire many-body system reacts collectively, the estimation precision scales super-extensively (e.g., Heisenberg scaling $\propto N^2$ or better, rather than standard shot-noise $\propto N$), significantly improving the signal-to-noise ratio for short interrogation times.
- **Collective Dissipation (Superradiance):** If multiple probes interact with the same thermal bath collectively rather than locally, they can exhibit Dicke-like superradiance. The collective coupling scales the effective dissipation rate by N, drastically cutting down the thermalization time while preserving a high QFI in the collective symmetric

subspace.

3. Ancilla-Assisted and Correlated Thermometry

These strategies involve entangling the probe with an auxiliary system that is explicitly shielded from the thermal bath.

- **Entanglement as a Buffer:** The probe interacts with the sample, but the measurement is performed jointly on the probe and the isolated ancilla. The metrological resource here is **initial entanglement** (such as two-mode squeezed vacuum states in continuous-variable thermometry). The isolated ancilla acts as a noise-free reference frame. This setup inherently bypasses the trade-off because you do not need the probe to fully thermalize; the transient degradation of the correlations encodes the temperature with extremely high precision at short times.

- **Continuous-Variable (CV) Squeezing:** In photonic or optomechanical thermometry, squeezed states are utilized. While thermalization generally destroys squeezing, tracking the rate of anti-squeezing or the variance in a specific quadrature provides a very fast temperature readout without needing the mechanical resonator or optical mode to reach thermal equilibrium.

4. Coarse-Grained and Thermodynamic Inference

Strict optimization of $\mathcal{F}_T$ assumes that you can perform complete, fine-grained projective measurements in the exact energy eigenbasis ($[H_P, \Pi_n] = 0$). In reality, doing this on complex probes takes too long and requires impossible control.

- **Maximum Entropy Inference:** For systems far from equilibrium, researchers are developing schemes that assign a "reference temperature" based on the Maximum Entropy principle constrained only by the mean energy of the non-equilibrium state. This allows for a direct, mathematically rigorous temperature readout without the system needing to be in a Gibbs state.

- **Coarse-Grained Measurements:** Tools from signal processing are being applied to show that optimal QFI can be closely approximated by binning energy levels rather than resolving them perfectly. The resource here is **algorithmic post-processing**—extracting maximum thermal information from suboptimal, easily accessible local observables (like local magnetization) at early times, rather than waiting for the global state to fully thermalize to measure total energy.